

Review Radar — design document

BIU 8688697201, Big Data and AI, final project (solo). Links rather than appendices: nothing here is restated from an ADR, a gate artefact or the [README](#).

1. The question, and who asks it

Fixed before any analysis ran ([ADR-0001](#)):

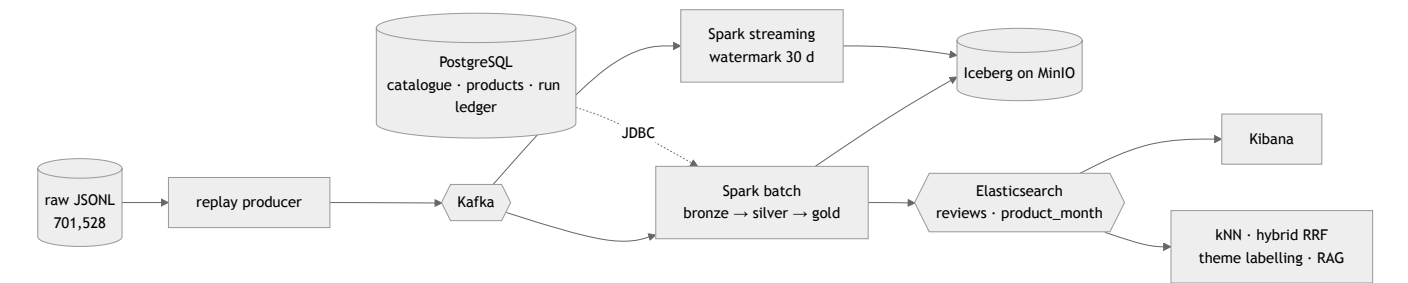
Which products experienced a sustained decline in customer ratings, and which complaint themes increased during that decline?

A category manager asks it — they own a shelf and want a shortlist, not a model. Amazon Reviews 2023, All_Beauty: **701,528 reviews**, **112,590 products**, **631,986 users**, **2000-11-01 → 2023-09-09**.

2. The data, and which V's it exercises

V	the measured figure
Volume	0.54 GB raw JSONL; 311 MB → 96.5 MB Parquet in bronze (3.2×); 693,547 ES docs; 345,418 vectors
Velocity	producer 232,000–484,000 rec/s — a range, because the host is shared and a point estimate would not reproduce; the P8 replay paces to a frozen 3,000 rec/s , measured 2,999
Variety	free text; a details dict whose keys collide case-insensitively; price as null / "9.99" / "\$9.99" / a range; products joined over JDBC
Veracity	(user, product, timestamp) collides in 6,139 groups over 13,415 rows — 6,138 byte-exact, 1 conflicting on helpful_vote, 0 disagreeing on rating → 7,276 removed

3. Architecture and data flow



PostgreSQL does three jobs: the **Iceberg catalogue** (the transactional pointer swap an object store cannot do, having no atomic rename), the **relational enrichment source** broadcast-joined into silver every run, and pipeline_runs, the **run ledger** whose run_id is stamped into every snapshot, ES doc and eval artefact ([ADR-0008](#)). **Exactly-once, as a figure:** make eos loads 120,000 records, SIGKILLS bronze mid-stream, restarts from the checkpoint — **89,964 rows before the kill, 120,000 after, 0 duplicate** (partition, offset) pairs ([transcript](#)).

4. The pipeline, phase by phase

One row per capability in `conf/lineage_chain.toml`, carrying what make eval-table prints today; tests/test_written_deliverables.py fails if the two disagree. PASS/FAIL is a *reproducibility* claim only — quality prints its own verdict and never reopens a phase ([ADR-0011](#)).

phase	capability	gate	verdict	the number, or the written reason
P2 Silver	silver	SILVER_GATE	PASS	8/8 · 701,528 → 694,252 rows, 0 rejects
P2 Silver	silver_repro	SILVER_REPRO_GATE	PASS	13/13 · re-derived in pandas, no shared code
P3 Gold	gold	GOLD_GATE	PASS	4/4 · 263,792 product-months, 425 alerts = 425 episodes, rule frozen
P3 Gold	gold_repro	GOLD_REPRO_GATE	PASS	11/11 · 0 mismatched fields over 263,792 + 167,518 + 425 rows
P3 Gold	gold_calibration	GOLD_CALIBRATION	FAIL	power 0.186 [0.168, 0.205] at the 0.3★ step vs bar 0.80, at 0.80 placebo alerts/month under a ceiling of 1.0 — §7
P3 Gold	gold_analytical	GOLD_ANALYTICAL	REPORTED	139 of 502 eligible products alerted in 2020–2023 (0.2769), 148 holdout episodes
P4 Search	search	SEARCH_GATE	PASS	7/7 · 693,547 + 263,792 docs, 7/7 analyzer cases
P5 Embeddings	embeddings	EMBED_GATE	PASS	7/7 · 345,418 vectors, ANN recall@10 0.96
P6 Themes	themes	THEMES_GATE	PASS	7/7 · taxonomy frozen, audit set opened once, 200/200
P6 Themes	themes_quality	THEMES_QUALITY	FAIL	macro-F1 0.4583 vs bar 0.70 — §7
P6 Themes	themes_agreement	THEMES_AGREEMENT	NOT_RUN	cut: Philip declined the hand-labelling pass (2026-09-14); two agent passes beside it, kappa 0.926 / 0.864, same model family — §7
P6 Themes	themes_repeat_kappa	THEMES_REPEAT_KAPPA	NOT_RUN	cut: a deterministic labeller has no repeat kappa
P7 RAG	rag	RAG_GATE	PASS	8/8 · 30/30 answered; 29/30 clear the reported contract
P7 RAG	rag_quality	RAG_QUALITY	NOT_RUN	cut: Philip declined the judging pass (2026-09-14); one agent pass beside it, 2 of 4 targets over their bars — §7
P8 Stream	stream_control	STREAM_GATE	PASS	11/11 · 129,330 product-months vs gold, 0 differing
P8 Stream	stream_demo	STREAM_GATE	PASS	18/18 · 2,000 far-slice rows dropped exactly, 882 diffs all explained

phase	capability	gate	verdict	the number, or the written reason
Lineage track	lineage	LINEAGE_GATE	PASS	7/7 · chain_clean=true over 101 links
Deliverables track	demo	DEMO_GATE	PASS	5/5 · 4 rehearsals (2 required), 259 s of 300 s, 4/4 documents

A capability is either a number or a written reason; the command exits non-zero on one that is neither. Every row above is now one or the other, so the command exits 0; the three NOT_RUN rows are cut, not pending, and each carries its reason.

5. The AI capability, and why this one

The course names the gap in deck 5: BM25 does not consider the semantic meaning of the query terms or the documents. P5 measures it in the same engine — **345,418** MiniLM-L6-v2 vectors over the ≥ 20 -word cohort, identity hashed over meaning-changing settings only ([ADR-0005](#)), indexed beside BM25, with hybrid fusing the two rankings client-side by RRF. **ANN recall@10 0.96; H-E1 is not held**, H-E2 and H-E3 are ([decision](#)). Above it: a frozen ten-theme **complaint** taxonomy — a theme set per review, not a polarity ([ADR-0003](#)) — and **RAG** with claim-level citations over the same retriever ([ADR-0006](#)), labelled by local qwen3:8b because ANTHROPIC_API_KEY is empty.

6. Results and insights

139 of 502 eligible products raised at least one sustained-decline alert during 2020–2023, under a rule developed on pre-2020 data and then applied unchanged — 148 holdout alerts, 148 holdout episodes, 27.7% of the eligible population. The rule was sealed in one commit (conf/decline_rule.toml, status = "frozen") *before* a single holdout point was evaluated, and the calibration that chose its thresholds is the commit before that. Over the whole spine: 263,792 product-months across 5,966 materialised products, 167,518 evaluation points of which 12,109 are evaluable, 425 alerts and 425 episodes, closed by recovery 207, gap 165, end of data 53.

The alert rate is the number; the detection rate is the caveat. The same calibration that licensed the freeze also measured what the rule misses: **power 0.186 [0.168, 0.205]** against a pre-registered bar of 0.80, at a placebo rate of **0.80 alerts/month** under a ceiling of 1.0 ([ADR-0012](#)). So an alert here is unlikely to be noise, and the 139 products are a floor rather than a census. *The rule finds declining products* does not follow from this and is not claimed. Worked example (demo 7): B00RPJZMUM, an ionic hair dryer at decline rank 3, where hard_to_use and not_as_described in 2016-10...2017-03 give way to does_not_work ("*motor stopped working*") in 2017-04...2017-09. **An alert is not causal proof**, and a shift over 18 baseline and 3 recent labelled mentions illustrates; it does not estimate.

7. How we avoided fooling ourselves

A **temporal holdout and protocol freeze** enforced in code, not left to the analyst; **reproducibility separated from outcome**, with two gates re-deriving the tables in pandas importing nothing from the Spark jobs (13 and 11 checks, 0 mismatched fields); **bars set before the measurement and not moved** — when the labeller dropped from hosted Haiku to a local 8B model, 0.70 stayed 0.70; and **every number joined back to a run**, make gate-lineage printing how many links it walked (chain_links_checked=101) so a pass over an empty chain is impossible.

P6's outcome, stated as the miss it is. On the once-opened, held-out audit set of 200 reviews the labeller scores **macro-F1 0.4583 [0.3647, 0.5223]** against **0.70** → THEMES_QUALITY=FAIL, blocking nothing. Scored on the same pass: the MLlib classifier **0.3899 [0.3074, 0.4639]** and the **star-only floor 0.2968 [0.2612, 0.3493]**. The labeller's interval and the floor's are **disjoint** — distinguishably better than predicting themes from the star rating — whereas the **classifier's overlaps the floor** and is not. On development the labeller and the floor overlapped; the holdout *reverses* that smaller-sample reading rather than confirming it. 27 of 200 labellings failed to parse (13.5%), each scoring as an empty prediction, so part of the gap precedes any judgement about themes.

Two human passes declined, and the rows cut rather than filled. THEMES_AGREEMENT was defined as the agent's ground truth against Philip's blind 50, and RAG_QUALITY as Philip's four-target judgement over the sealed thirty. Philip declined both on 2026-09-14, so both capabilities are **cut** in conf/lineage_chain.toml with the reason printed in the table, and the correlated-error risk of machine-made ground truth is now **unpriced** rather than priced. The agent passes that exist beside them are published as what they are: two same-model-family consistency checks on the 50 (kappa 0.926 and 0.864, reproducibility not accuracy) and one different-model-family pre-check on the thirty (grounded 11/20 and adequate 10/20 under their bars, abstention 10/10 and false refusal 1/20 over theirs). No agent label is presented as human, and no bar moved (ADR-0003 amendment e, ADR-0006).

8. Trade-offs, and what was declined out loud

Three trade-offs. Iceberg is not a course technology — one mention in ~860 slides, conceded up front. **Local qwen3:8b instead of hosted Haiku**, with the 0.70 bar deliberately unmoved, so the constraint surfaces as a reported miss rather than a relaxed target. **Public but sensitive review text** — only title and text cross the wire to the model, never user_ids.

Four declines, in the wording that binds them, and a fifth internal to the lineage gate.

- **Kafka Connect Elasticsearch sink — dropped for time.** Not argued from the deck's Except for a trivial "file" connector line, which covers a Connect *source*; the deck separately names an Elasticsearch sink. The schedule decided this, not the merits.
- **Single-node HDFS — declined on resources, conceding that MinIO is not equivalent:** block replication and rack awareness against a flat key space with no atomic rename.
- **Sqoop and Pig — on the course's own slide:** Apache Sqoop moved into the Attic in June 2021; Spark SQL does Pig's transform.
- **Oozie — declined as an opinion, and owned as one.** Spark subsumes the orchestration and the run ledger fills the role, but Oozie is not retired and the decks teach it across 25 mentions. GraphFrames/GraphX go the same way, as a detour.
- **The lineage gate's publication mode — declined as unreachable, not as pending.** The walk pins superseded runs on purpose (P6's frozen frames, the pre-freeze calibration run, P8's two projections), so publication_ready cannot print true in any run order; the claim is chain_clean=true over chain_links_checked=101 ([ADR-0013](#)).